

The cross-section $\mathbf{A} \rightarrow \mathbf{A}'$ (§1.544)

Freyd & Scedrov, §1.544 — the inflation and its section

A **cross-section** of the forgetful functor $\text{listProd} : \mathbf{A}' \rightarrow \mathbf{A}$ is a functor going the other way, $\mathbf{A} \rightarrow \mathbf{A}'$, that listProd undoes. The forgetful functor is **many-to-one**: it sends a list s to its product $\prod s$, and many different lists have the same product (up to isomorphism), so many lists sit over each object A . The cross-section chooses one of them — the one-element list $[A]$.

$\mathbf{A}' = \text{List } \mathcal{C}$ — objects are lists, product = concatenation

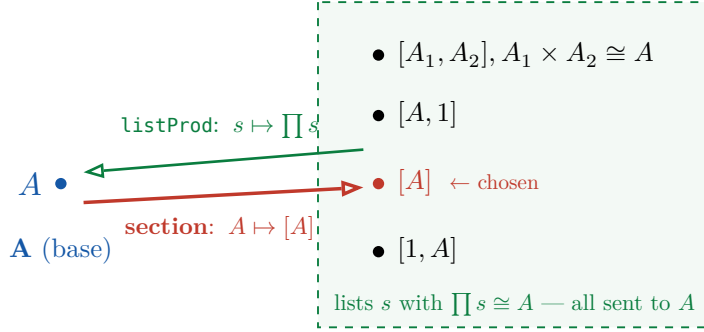


Figure 1: listProd (green) sends every list over A down to A (it is many-to-one). The cross-section (red) lifts A back up to the **single** list $[A]$. It is a **section** because going up then down returns to A : $A \mapsto [A] \mapsto \prod [A] = A$, i.e. $\text{listProd} \circ \text{section} = \text{id}$.

That round trip is the defining property — drawn as a triangle:

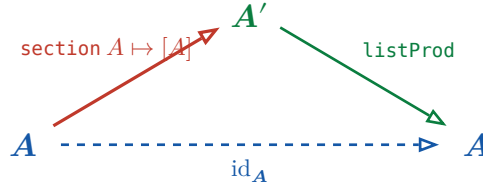


Figure 2: $\text{listProd} \circ \text{section} = \text{id}$: the cross-section is a one-sided (right) inverse of the forgetful functor.

Why §1.544 wants it. The slice inclusion $\mathbf{A} \rightarrow \mathbf{A}/B$ is split into two strict steps:

$$\mathbf{A} \xrightarrow{A \mapsto [A]} \mathbf{A}' \xrightarrow{s \mapsto s ++ [B]} \mathbf{A}'/B$$

The cross-section is the first arrow — it relabels each object A of $\mathbf{A}$ as the singleton list $[A]$, so the second step (appendFunctor B , “multiply by B ” done as list concatenation) has lists to append to.